

COMPANION
to

ChatGPT Teaches TikZ

Chapter 24: Layers and Visual Structure (A)

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24.1 What This Companion Adds

Chapter 24 separated source order from painting order. This Companion develops a three-level visual architecture: fitted regions behind the mathematics, the main diagram in the ordinary layer, and selected annotations in a foreground layer.

Design

Use layers to record visual hierarchy explicitly. Do not distort the logical order of the source merely to make a background paint first.

24.2 A Layer Vocabulary

Command or option	Purpose
<code>\pgfdeclarelayer</code>	Declares a named layer.
<code>\pgfsetlayers</code>	Sets the complete back-to-front order.
<code>pgfonlayer</code>	Places commands on an explicit layer.
<code>on background layer</code>	Places a scope on the standard background layer.
<code>show background rectangle</code>	Adds a rectangle around the whole picture.
<code>background rectangle/.style</code>	Styles the whole-picture background.
<code>fit=(A) (B)</code>	Makes a node enclose named objects.
<code>inner sep</code>	Adds space inside a fitted boundary.

Load the convenience libraries with

```
\usetikzlibrary{backgrounds,fit}
```

24.3 Why Source Order Is Not Enough

TikZ paints later objects over earlier ones. A rectangle drawn after two nodes can cover them.

```
\node[draw] (A) at (0,0) {$A$};  
\node[draw] (B) at (2,0) {$B$};  
\fill[gray!30] (-.5,-.5) rectangle (2.5,.5);
```

Moving the fill command earlier fixes the painting order but can make the source harder to organize, especially when the rectangle depends on the nodes.

24.4 Declaring Explicit Layers

```
\pgfdeclarelayer{background}  
\pgfdeclarelayer{foreground}  
\pgfsetlayers{background,main,foreground}
```

The order is back to front. The ordinary TikZ layer is named `main`.

Common Mistake

The `\pgfsetlayers` command states the complete active order. Do not omit `main` unless you intentionally want to remove the ordinary layer.

24.5 Drawing on a Named Layer

```
\begin{pgfonlayer}{background}  
  \fill[gray!20] (-.5,-.5) rectangle (2.5,.5);  
\end{pgfonlayer}
```

This block may appear after the nodes in the source; it is still painted before the main layer.

24.6 The Background Convenience Scope

For an ordinary background, the library provides a shorter form.

```
\begin{scope}[on background layer]  
  \fill[gray!20] (-.5,-.5) rectangle (2.5,.5);  
\end{scope}
```

Use explicit layers when the picture needs more than one controlled level or when foreground annotations must be guaranteed to remain on top.

24.7 A Background for the Entire Picture

```
\begin{tikzpicture}[
  show background rectangle,
  background rectangle/.style={
    draw,rounded corners,fill=gray!10
  }
]
...
\end{tikzpicture}
```

This rectangle follows the picture's complete bounding box. It is appropriate for a panel, not for grouping only selected nodes.

24.8 Fitting a Region to Named Objects

```
\node[
  draw,rounded corners,fit=(A)(B)(C),inner sep=5mm
] {};
```

The fitted node adapts when its contents move or their labels change.

```
\begin{scope}[on background layer]
  \node[subsystem,fit=(A)(B)(C)] {};
```

The fitting calculation occurs after the objects are positioned, while the layer keeps the region behind them.

24.9 Backgrounds Must Not Control Geometry

Position the mathematical nodes first. Let fitted regions follow those nodes. If foreground objects are positioned relative to a background rectangle, a later change in decoration can accidentally change the mathematics.

24.10 Worked Project: A Structured Graph

24.10.1 Stage 1: Define the Visual Roles

```
\tikzset{
  vertex/.style={draw,circle,fill=white,minimum size=7mm},
  edge/.style={thick},
  subsystem/.style={draw,rounded corners,
    fill=gray!10,inner sep=5mm},
  emphasis/.style={very thick,-{Stealth}}
}
```

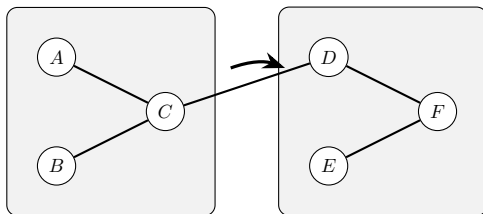
24.10.2 Stage 2: Draw the Main Graph

```
\node[vertex] (A) at (0,1) {$A$};
\node[vertex] (B) at (0,-1) {$B$};
\node[vertex] (C) at (2,0) {$C$};
\node[vertex] (D) at (5,1) {$D$};
\node[vertex] (E) at (5,-1) {$E$};
\node[vertex] (F) at (7,0) {$F$};
\draw[edge] (A)--(C)--(B) (D)--(F)--(E) (C)--(D);
```

24.10.3 The Complete Picture

```
\begin{tikzpicture}
  \pgfdeclarelayer{foreground}
  \pgfsetlayers{background,main,foreground}
  % Main nodes and edges are defined here.
  \begin{scope}[on background layer]
    \node[subsystem,fit=(A)(B)(C)] {};
    \node[subsystem,fit=(D)(E)(F)] {};
  \end{scope}
  \begin{pgfonlayer}{foreground}
    \draw[emphasis] (3.2,.8) to[bend left=20] (4.2,.8);
  \end{pgfonlayer}
\end{tikzpicture}
```

See



Design

Fitted regions record conceptual groups, the main layer records the graph, and the foreground arrow records emphasis. Each visual responsibility is explicit.

24.11 Debugging Workshop

24.11.1 A Background Covers the Nodes

Move it to the background layer rather than moving its source command earlier.

24.11.2 A Fitted Region Does Not Follow a Node

Check that the node's name appears in the `fit` list and that the `fit` library is loaded.

24.11.3 A Foreground Annotation Disappears

Declare the foreground layer, include it after `main`, and place the annotation inside the correct `pgfonlayer` environment.

24.11.4 The Background Changes the Layout

Remove positioning dependencies on decorative regions. Main geometry should determine the background, not the reverse.

24.12 Exercises

1. Place a colored region behind three named nodes.
2. Move one node and verify that a fitted region follows.
3. Add a foreground annotation to a two-layer diagram.
4. Create a whole-picture background rectangle.

5. Compare a manual rectangle with a fitted node.
6. Define a semantic style for all conceptual regions.
7. Build a three-layer picture with explicit layer order.
8. Refactor a diagram whose backgrounds are drawn first manually.

24.13 Solutions

24.13.1 Exercises 1–4

```
\begin{scope}[on background layer]
  \node[fill=gray!15,fit=(A)(B)(C),inner sep=4mm] {};
\end{scope}
```

Exercise 2 changes only the main node position. Exercise 3 declares a foreground layer and places the annotation there.

24.13.2 Exercises 5–7

```
\pgfdeclarelayer{foreground}
\pgfsetlayers{background,main,foreground}
```

Exercise 5 demonstrates that `fit` adapts automatically. Exercise 6 moves all region appearance into one style. Exercise 7 assigns each object to the layer that matches its visual responsibility.

For Exercise 8, keep the logical source order and move only the painting order into the layer declarations.

24.14 ChatGPT Workshop

Ask ChatGPT to preserve geometry while reorganizing the visual hierarchy.

1. Identify which objects belong on background, main, and foreground layers.
2. Replace manual grouping rectangles with fitted background nodes.
3. Diagnose why this later background command covers the main diagram.
4. Check the complete layer order and every layer environment.
5. Ensure decorative regions follow, but never determine, the geometry.

24.15 Final Checklist

Before continuing to Chapter 25, check that you can

1. explain ordinary source-order painting;
2. declare and order explicit layers;
3. use the background convenience scope;
4. fit a region around named objects;
5. place annotations reliably in the foreground;
6. keep mathematical geometry independent of decoration; and
7. use layers to express visual responsibility.